# Assignment - MongoDB

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| **Reg NO** | : | 2021/ICT/59 |
| **Semester** | : | 2.2 |
| **Course Code** | : | IT-2234(P) |
| **Course Name** | : | Web Services and Server Technologies (Practical) |

Registration Number: 2021/ICT/59

# IT-2234(P) - Web Services and Server Technologies
# Department of Physical Science
# University of Vavuniya
# Assignment – MongoDB

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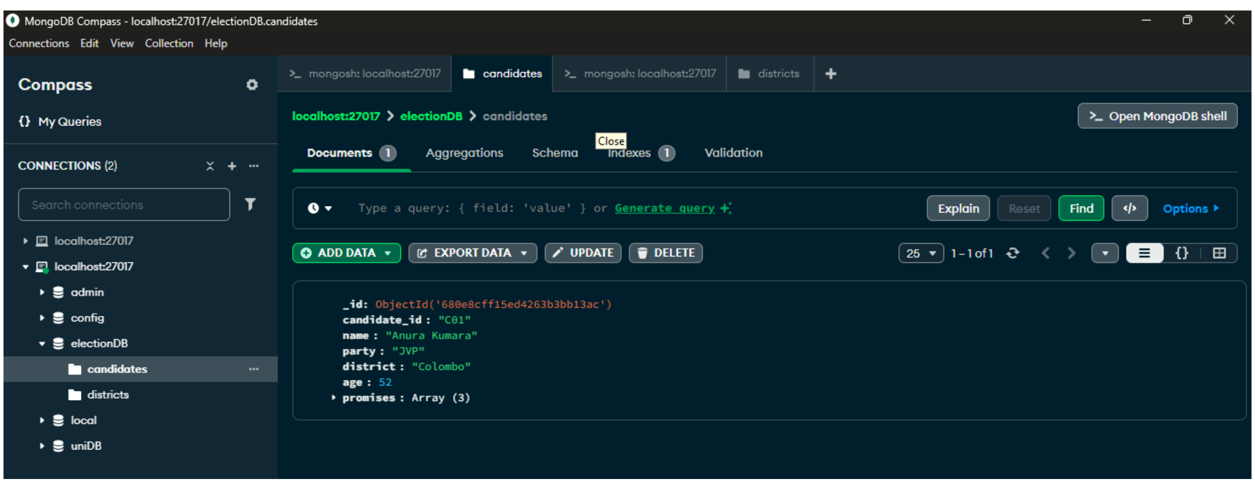
## Question 01

1. **Create a database, collection and insert data.**

   **Code:**

```
use electionDB

db.candidates.insertOne({
  "candidate_id": "C1001",
  "name": "Anura Kumara",
  "party": "JVP",
  "district": "Colombo",
  "age": 52,
  "promises": ["Free Education", "Healthcare", "Jobs"]
})
```




2. **Insert many candidates at once.**

   **Code:**

```
db.candidates.insertMany([
```

```
    {
       "candidate_id": "C02",
       "name": "Sajith Premadasa",
       "party": "SJB",
       "district": "Colombo",
       "age": 56,
       "promises": ["Housing", "Education"]
    },

    {
       "candidate_id": "C03",
       "name": "Gotabaya Rajapaksa",
       "party": "SLPP",
       "district": "Kandy",
       "age": 75,
       "promises": ["Security", "Infrastructure"]
    },

    {
       "candidate_id": "C04",
       "name": "Ruwan Wijewardene",
       "party": "UNP",
       "district": "Gampaha",
       "age": 48,
       "promises": ["Youth power", "Technology", "Education"]
     }

])
```

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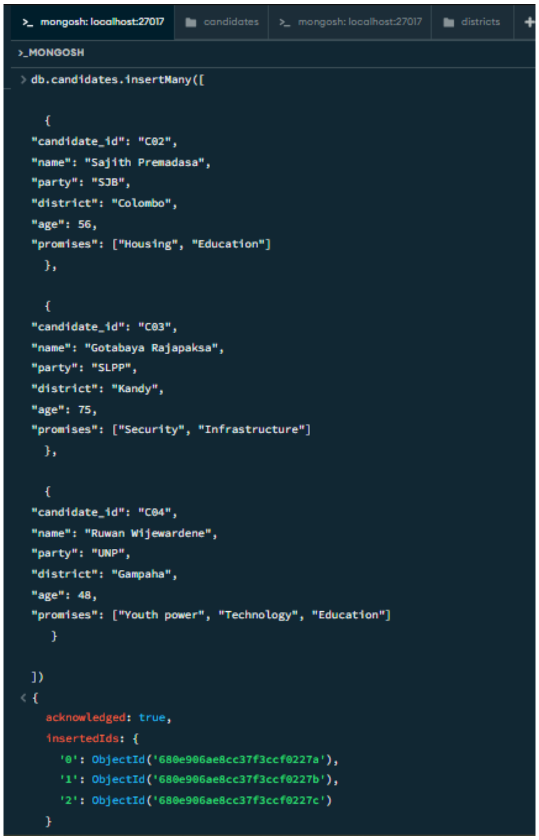
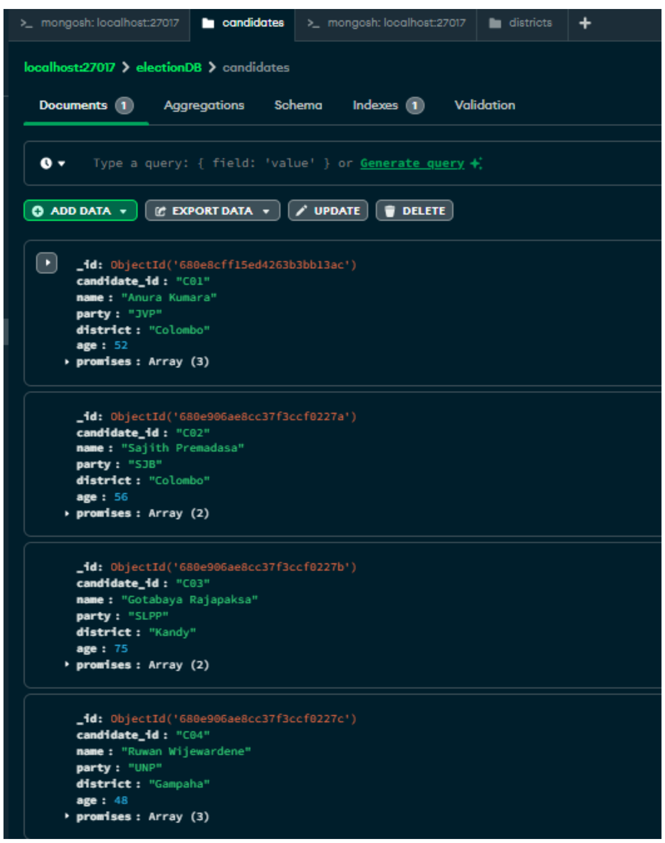
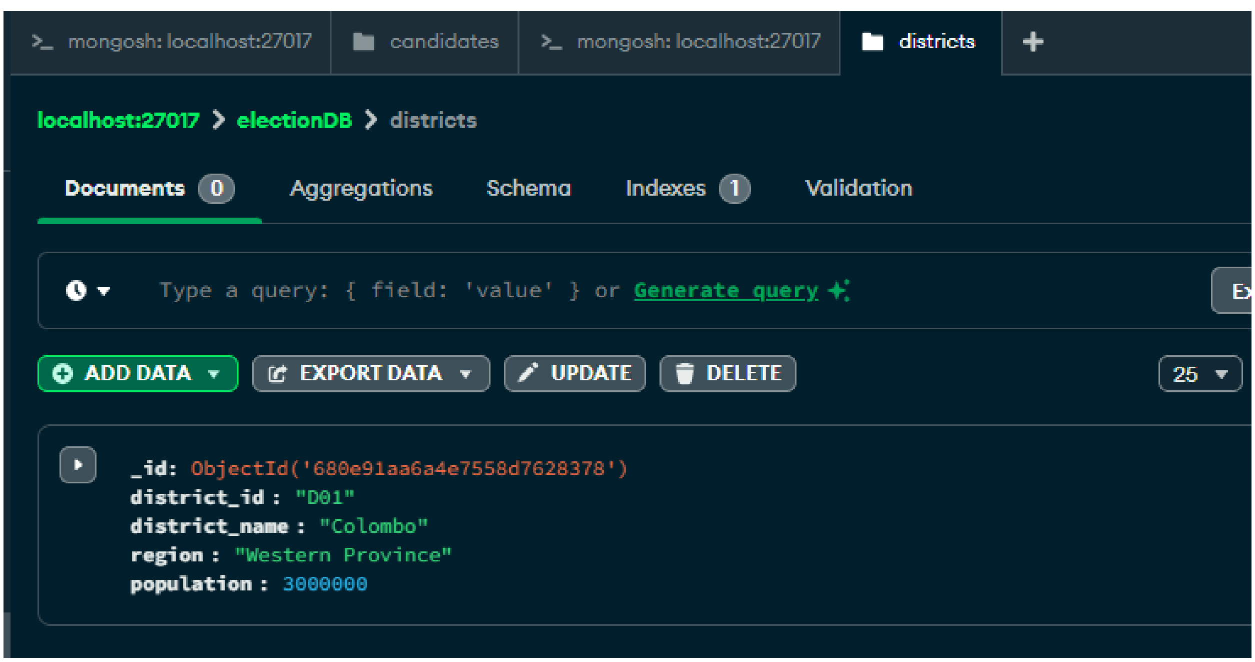


**3. Create a districts collection and insert data.**

**Code:**

```
db.districts.insertOne({
  "district_id": "D01",
  "district_name": "Colombo",
  "region": "Western Province",
  "population": 3000000
})
```

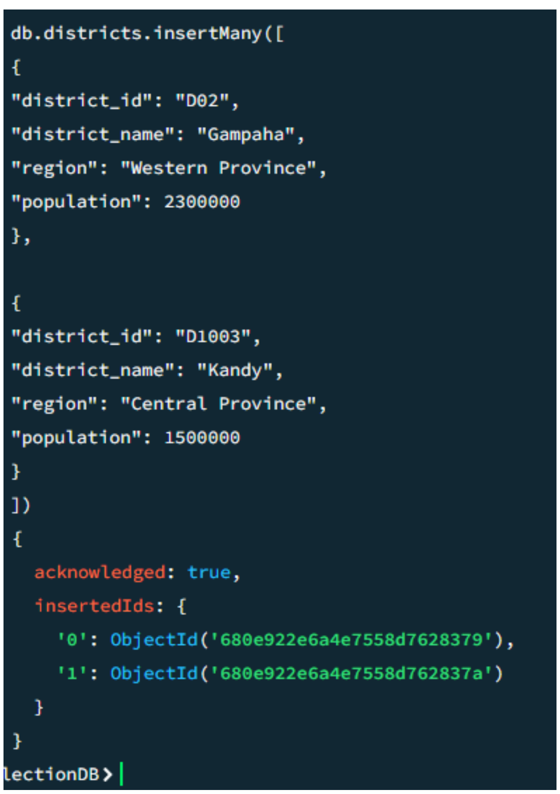



4. **Insert many districts at once.**

**Query:**

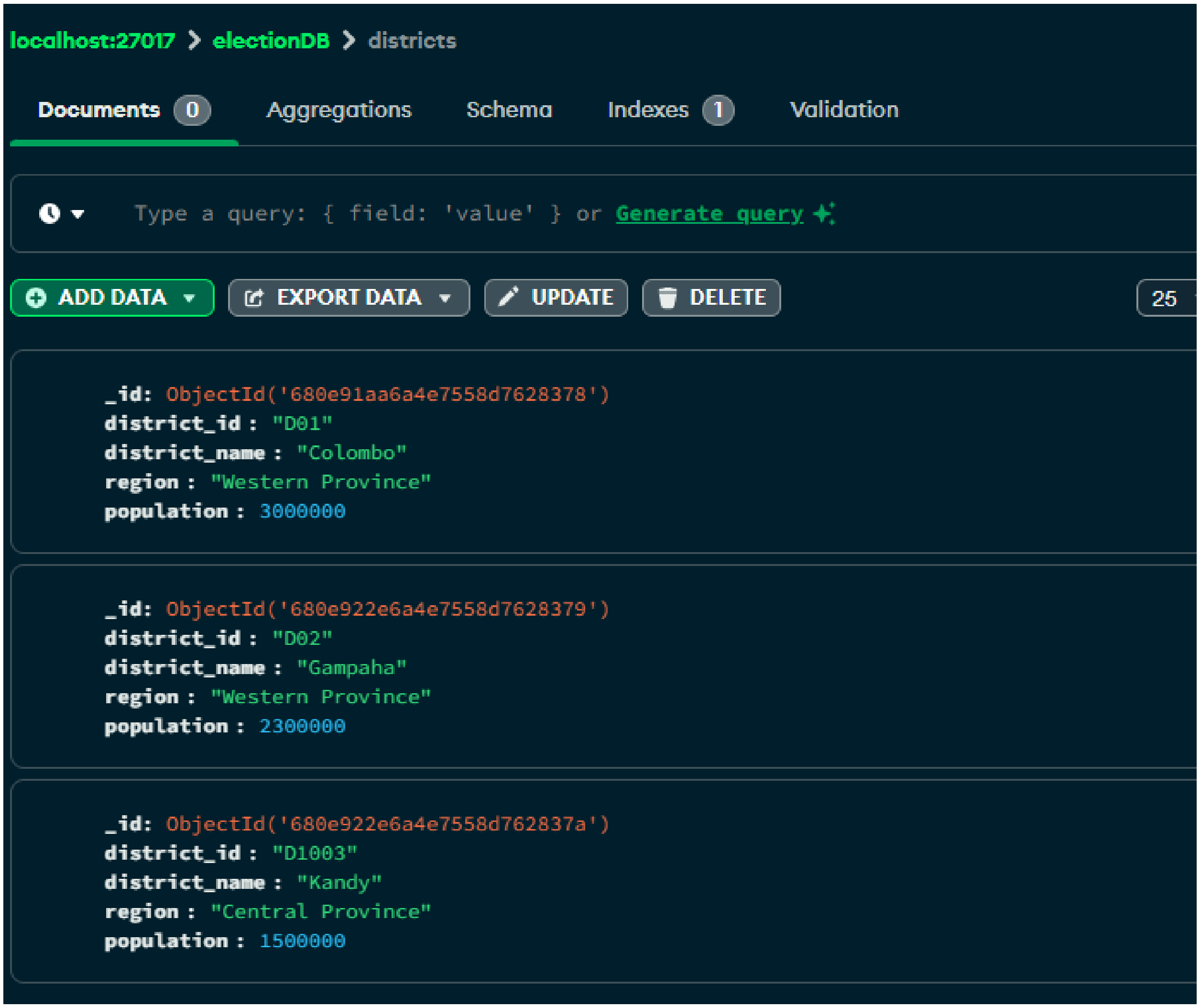
```
db.districts.insertMany([
{
"district_id": "D02",
"district_name": "Gampaha",
"region": "Western Province",
"population": 2300000
},

{
"district_id": "D1003",
"district_name": "Kandy",
"region": "Central Province",
"population": 1500000
}
])
```



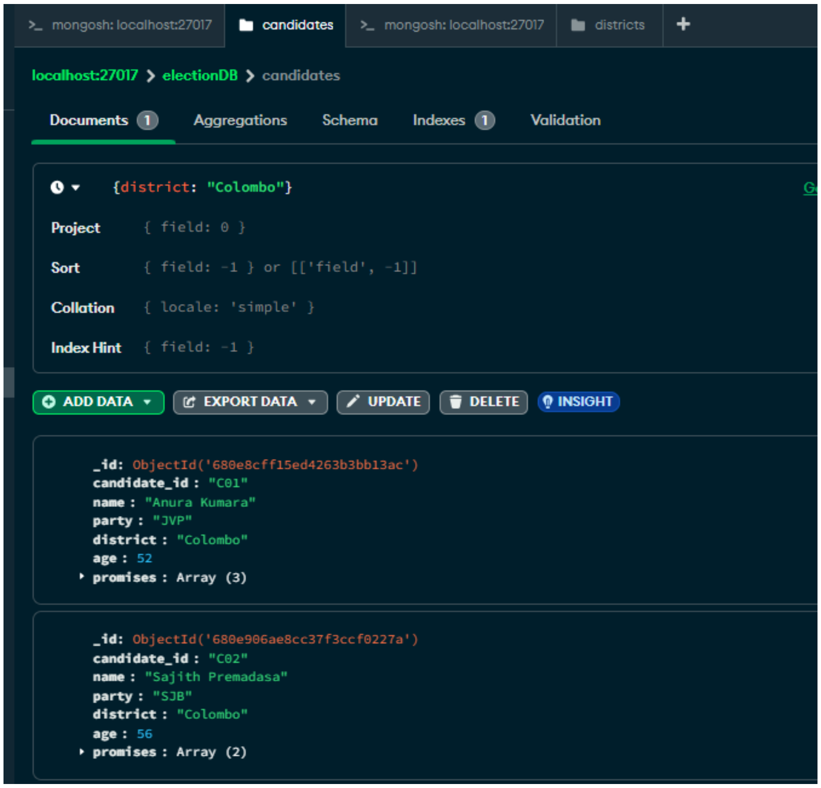
```
db.districts.insertMany([
{
"district_id": "D02",
"district_name": "Gampaha",
"region": "Western Province",
"population": 2300000
},

{
"district_id": "D1003",
"district_name": "Kandy",
"region": "Central Province",
"population": 1500000
}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('680e922e6a4e7558d7628379'),
    '1': ObjectId('680e922e6a4e7558d762837a')
  }
}
lectionDB>
```




5. **List all candidates from the Colombo district**

   **Query:**
   ```
   {district: "Colombo"}
   ```



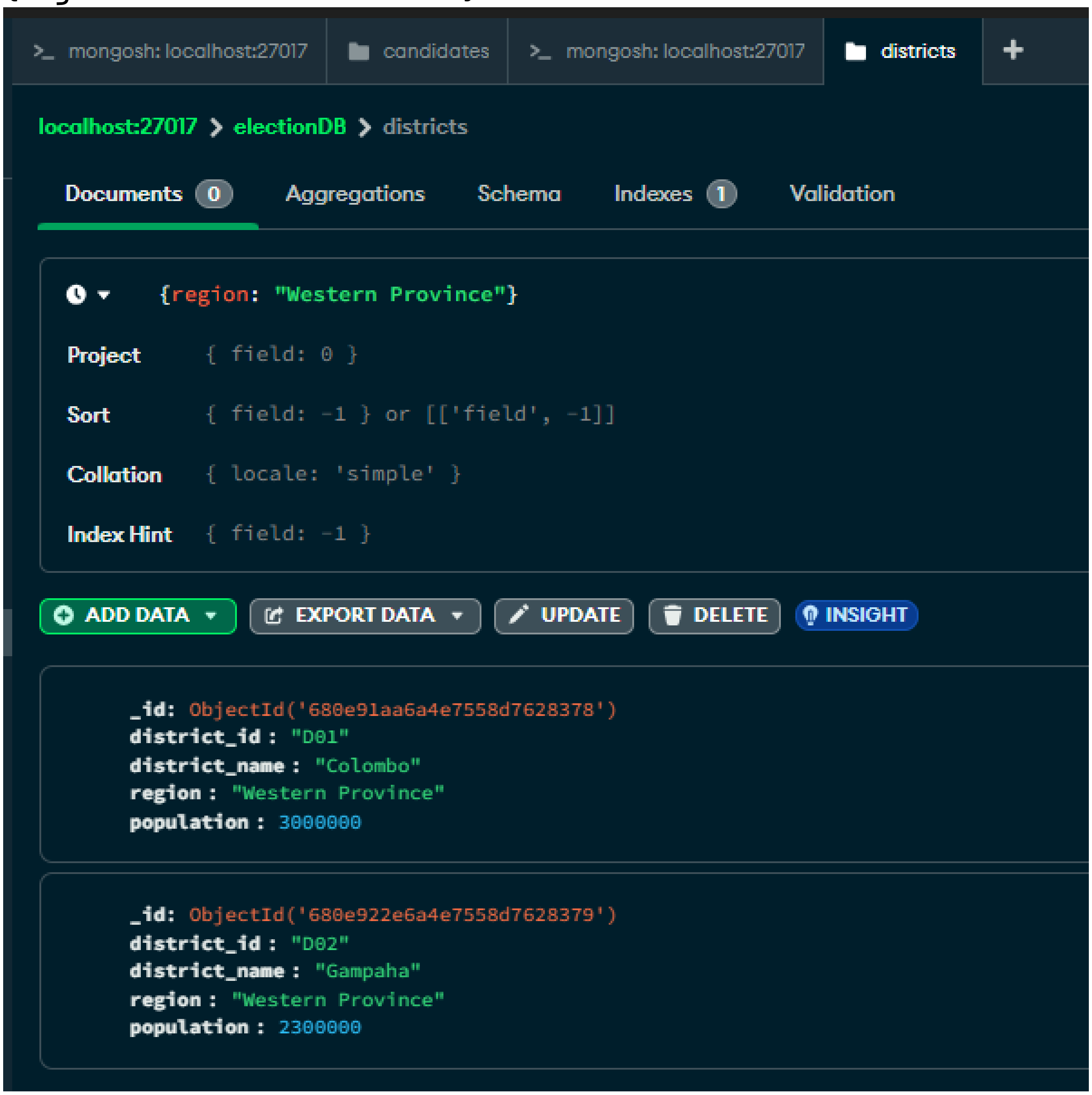

**Shell Query:**

```
db.candidates.find({ "district": "Colombo" })
```

```
> db.candidates.find({ "district": "Colombo" })
< {
    _id: ObjectId('680e8cff15ed4263b3bb13ac'),
    candidate_id: 'C01',
    name: 'Anura Kumara',
    party: 'JVP',
    district: 'Colombo',
    age: 52,
    promises: [
      'Free Education',
      'Healthcare',
      'Jobs'
    ]
  }
  {
    _id: ObjectId('680e906ae8cc37f3ccf0227a'),
    candidate_id: 'C02',
    name: 'Sajith Premadasa',
    party: 'SJB',
    district: 'Colombo',
    age: 56,
    promises: [
      'Housing',
      'Education'
    ]
  }
electionDB>
```

6. **Find all districts in the Western Province.**

**Query:**
`{region: "Western Province"}`

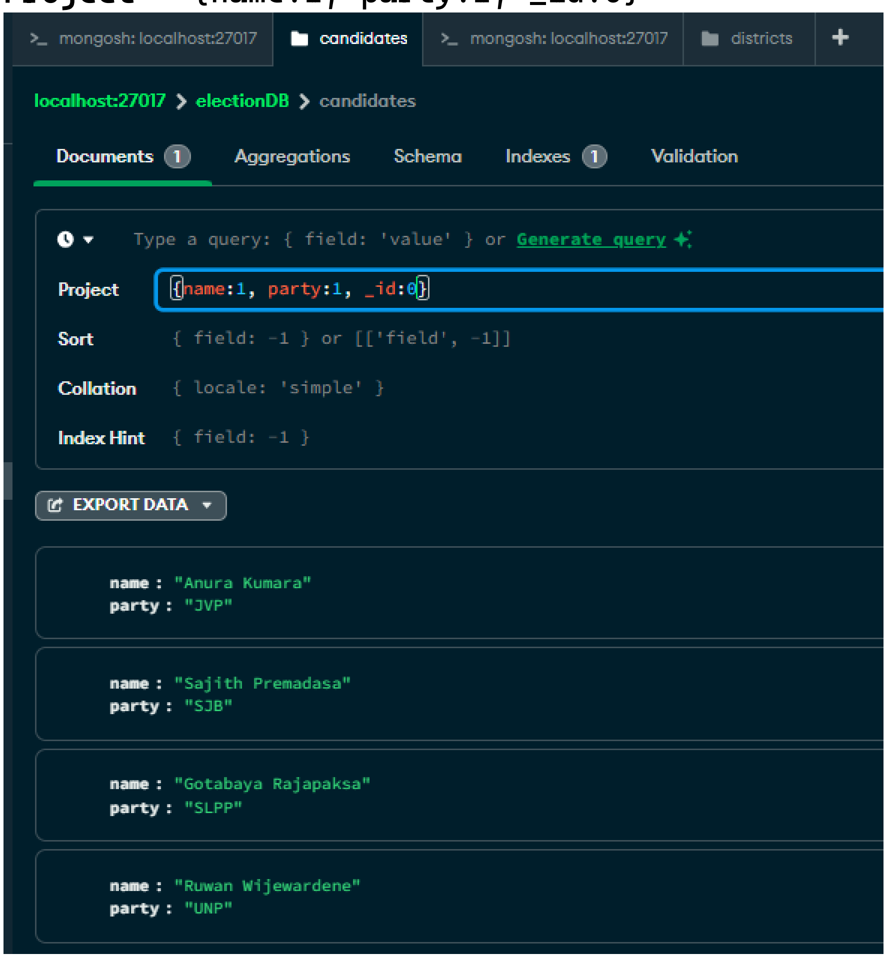



**Shell Query:**
```
db.districts.find({ "region": "Western Province" })
```


7. **Show only the name and party of all candidates.**

**Query:**

```
Project → {name:1, party:1, _id:0}
```



**Shell Query:**

```
db.candidates.find({}, {name:1, party:1, _id:0})
```

```
> db.candidates.find({}, {name:1, party:1, _id:0})
< {
    name: 'Anura Kumara',
    party: 'JVP'
  }
  {
    name: 'Sajith Premadasa',
    party: 'SJB'
  }
  {
    name: 'Gotabaya Rajapaksa',
    party: 'SLPP'
  }
  {
    name: 'Ruwan Wijewardene',
    party: 'UNP'
  }
electionDB>
```

8. **Find the party of the candidate named "Anura Kumara".**

**Query:**

```
db.candidates.find({ "name": "Anura Kumara" }, { "party": 1, _id: 0 })
```

```
> db.candidates.find({ "name": "Anura Kumara" }, { "party": 1, _id: 0 })
< {
    party: 'JVP'
  }
electionDB>
```

9. **Find candidates who are older than 55 years.**

**Query:**

```
{age: {$gt: 55}}
```

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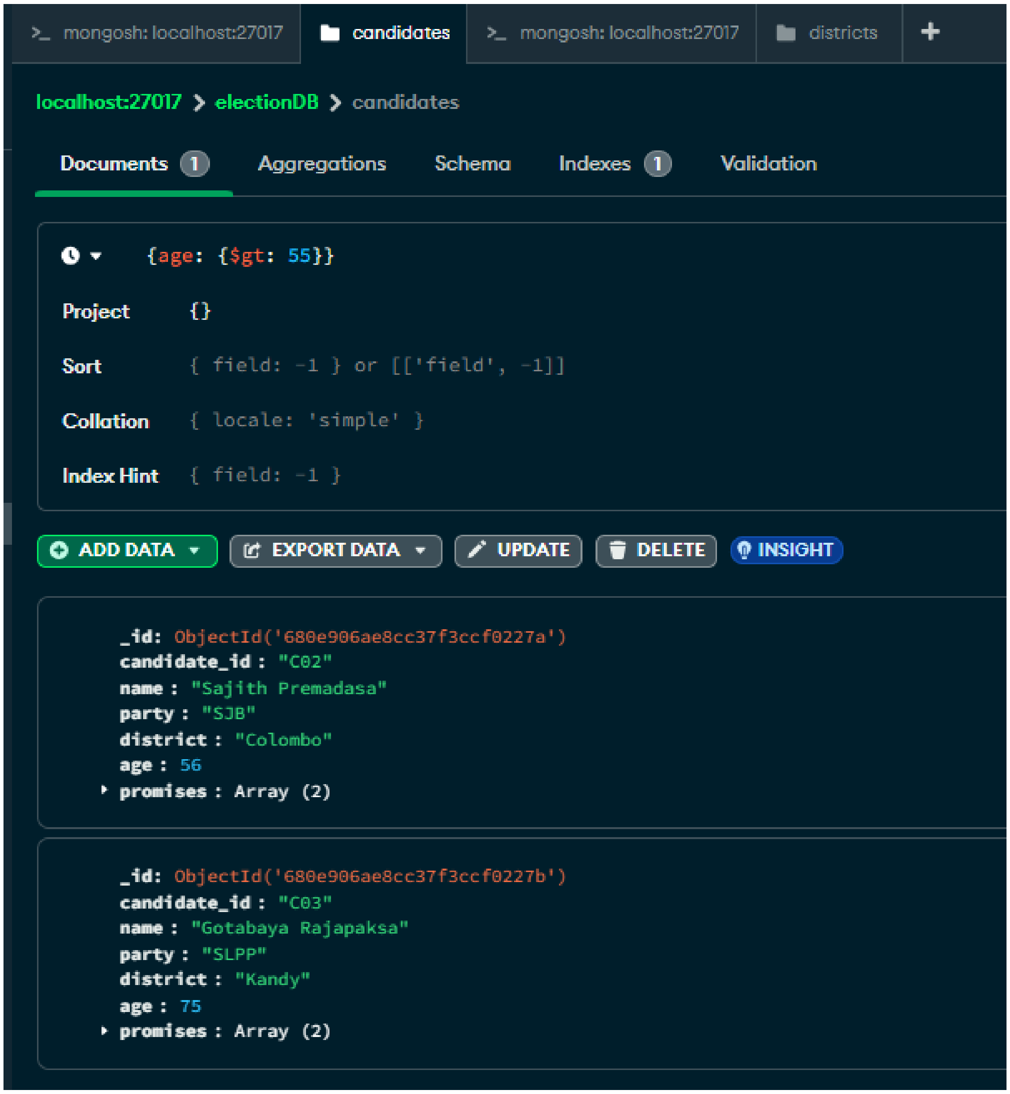


**Shell Query:**

```
db.candidates.find({age: {$gt: 55}})
```

```
> db.candidates.find({age: {$gt: 55}})
< {
    _id: ObjectId('680e906ae8cc37f3ccf0227a'),
    candidate_id: 'C02',
    name: 'Sajith Premadasa',
    party: 'SJB',
    district: 'Colombo',
    age: 56,
    promises: [
      'Housing',
      'Education'
    ]
  }
  {
    _id: ObjectId('680e906ae8cc37f3ccf0227b'),
    candidate_id: 'C03',
    name: 'Gotabaya Rajapaksa',
    party: 'SLPP',
    district: 'Kandy',
    age: 75,
    promises: [
      'Security',
      'Infrastructure'
    ]
  }
electionDB>
```

**10. Find the districts whose population is greater than 2 million**

**Query:**

```
{ "population": { $gt: 2000000 } }
```

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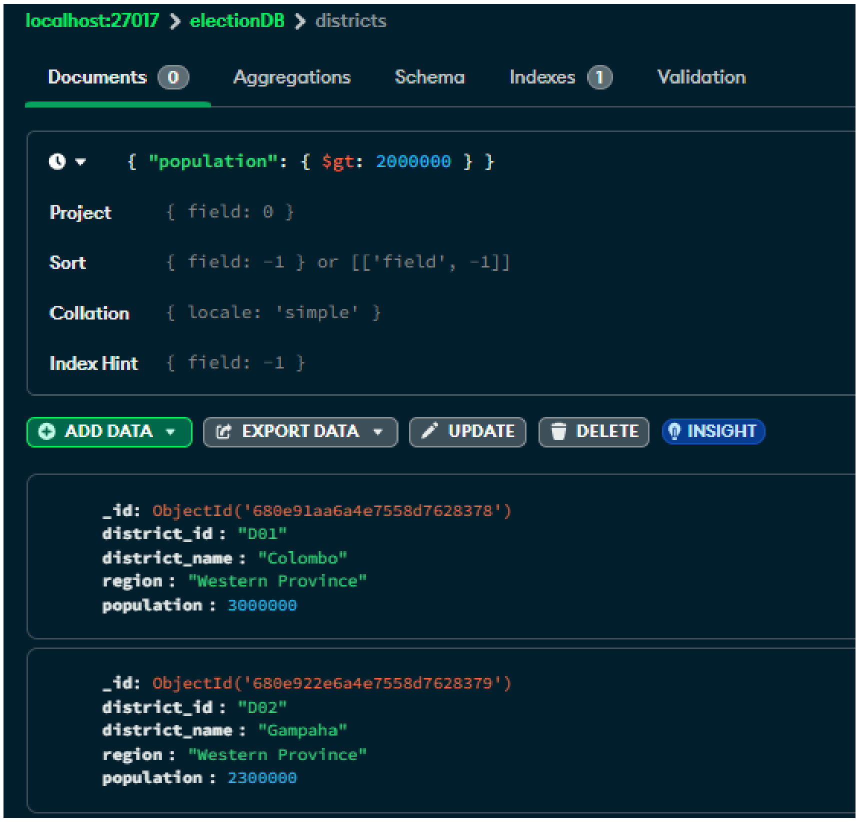



**Shell Query:**

```
db.districts.find({ "population": { $gt: 2000000 } })
```

**11. Find candidates who have "Education" mentioned in their promises.**

**Shell Query:**

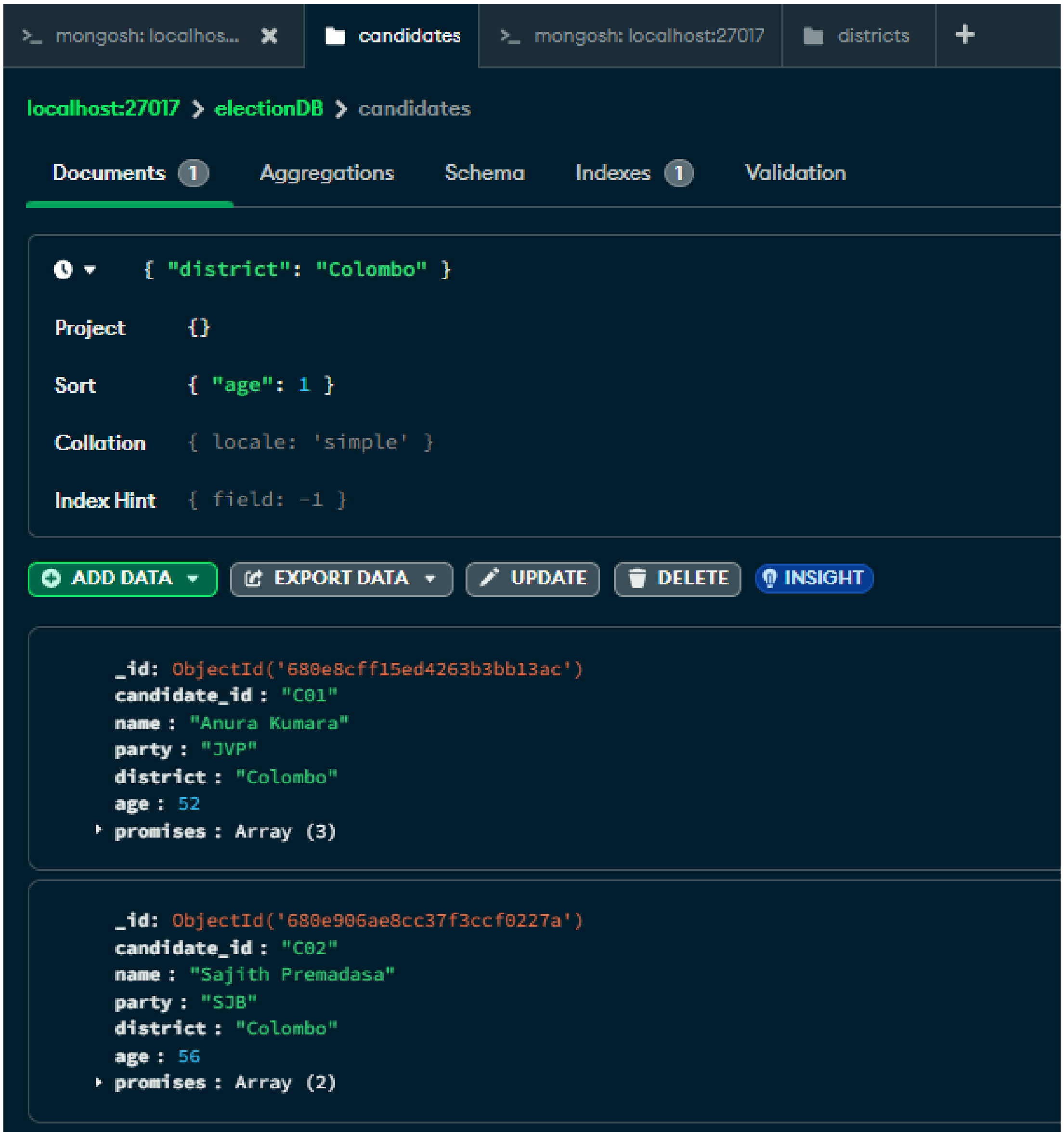
```
db.candidates.find({ "promises": { $in: ["Education"] } })
```

```
> db.candidates.find({ "promises": { $in: ["Education"] } })
< {
    _id: ObjectId('680e906ae8cc37f3ccf0227a'),
    candidate_id: 'C02',
    name: 'Sajith Premadasa',
    party: 'SJB',
    district: 'Colombo',
    age: 56,
    promises: [
      'Housing',
      'Education'
    ]
  }
  {
    _id: ObjectId('680e906ae8cc37f3ccf0227c'),
    candidate_id: 'C04',
    name: 'Ruwan Wijewardene',
    party: 'UNP',
    district: 'Gampaha',
    age: 48,
    promises: [
      'Youth power',
      'Technology',
      'Education'
    ]
  }
electionDB>
```

**12. Find candidates from Colombo sorted by age (ascending).**

**Query:**

```
Filter -> { "district": "Colombo" }
Sort   -> { "age": 1 }
```




**Shell Query:**

```
db.candidates.find({ "district": "Colombo" }).sort({ "age": 1 })
```

```
> db.candidates.find({ "district": "Colombo" }).sort({ "age": 1 })
< {
    _id: ObjectId('680e8cff15ed4263b3bb13ac'),
    candidate_id: 'C01',
    name: 'Anura Kumara',
    party: 'JVP',
    district: 'Colombo',
    age: 52,
    promises: [
      'Free Education',
      'Healthcare',
      'Jobs'
    ]
  }
  {
    _id: ObjectId('680e906ae8cc37f3ccf0227a'),
    candidate_id: 'C02',
    name: 'Sajith Premadasa',
    party: 'SJB',
    district: 'Colombo',
    age: 56,
    promises: [
      'Housing',
      'Education'
    ]
  }
electionDB>
```

**13. Sort the districts by population.**

**Query:**

```
Sort   -> { "population": 1 }
```

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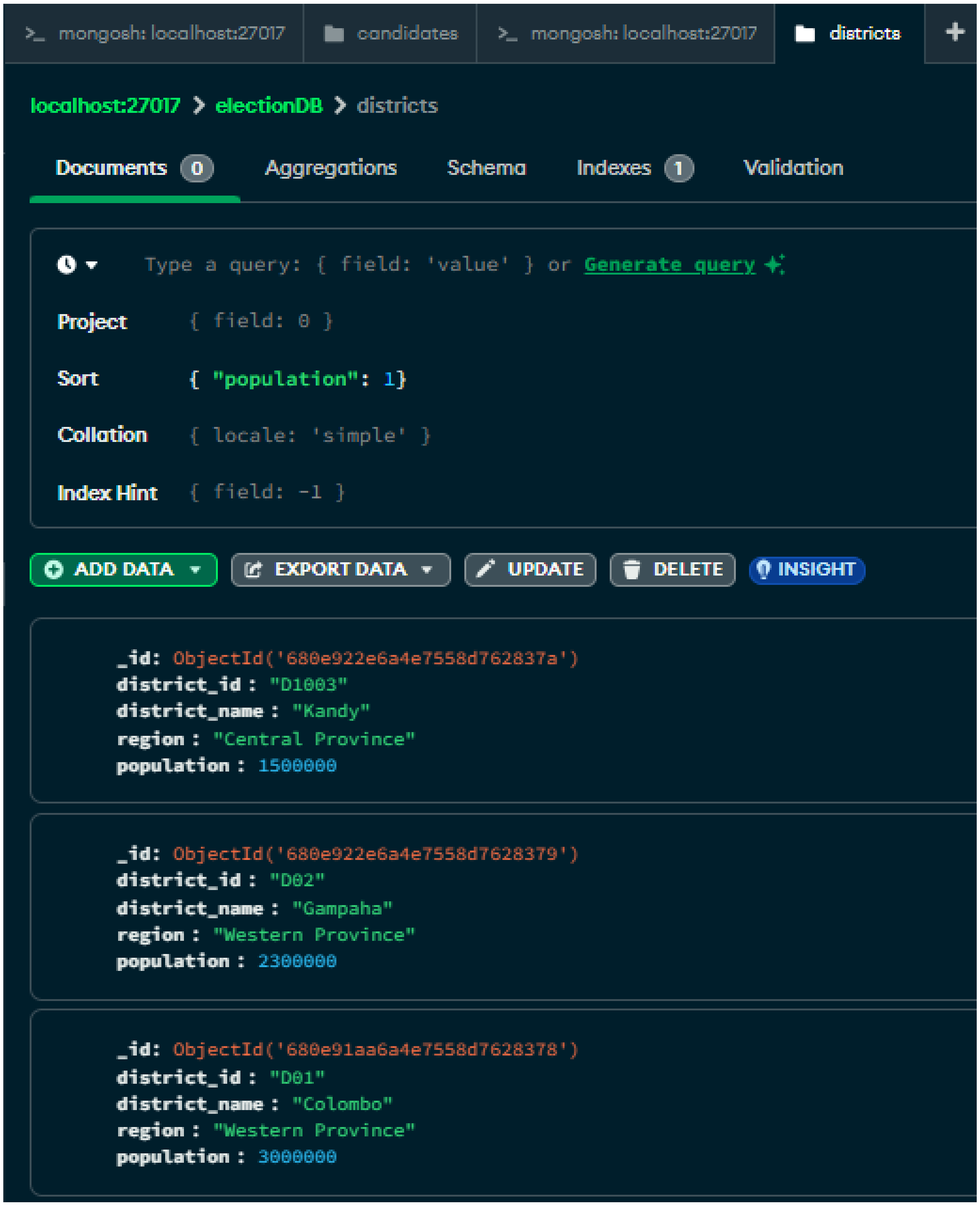

**Shell Query:**

```
db.districts.find().sort({ population:1 })
```

```
> db.districts.find().sort({ population:1 })
< {
    _id: ObjectId('680e922e6a4e7558d762837a'),
    district_id: 'D1003',
    district_name: 'Kandy',
    region: 'Central Province',
    population: 1500000
  }
  {
    _id: ObjectId('680e922e6a4e7558d7628379'),
    district_id: 'D02',
    district_name: 'Gampaha',
    region: 'Western Province',
    population: 2300000
  }
  {
    _id: ObjectId('680e91aa6a4e7558d7628378'),
    district_id: 'D01',
    district_name: 'Colombo',
    region: 'Western Province',
    population: 3000000
  }
electionDB>
```